

Sherif Osama

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PROFESSIONAL SUMMARY

Driven Electronics and Communications Engineering student with a solid foundation in Embedded Systems and IoT. Proven leadership experience as Head of Embedded Systems at IEEE Al-Azhar. Adept at C/C++ and Python programming, alongside microcontroller interfacing. Seeking an internship opportunity to leverage my technical skills and contribute to innovative embedded solutions.

EDUCATION

Al-Azhar University, *Bachelor of Electronics & Communications Engineering* 2027
Relevant Coursework: Microprocessors, Embedded Systems, Logic Design, C++ Programming.

TECHNICAL SKILLS

Languages (C, Embedded C, C++, Python, Assembly , Shell Scripting (POSIX sh, bash)) | **Microcontrollers** (AVR (ATmega32), PIC, STM32 (ARM Cortex-M), Arduino) | **Protocols** (USART, SPI, I2C, MQTT) | **Tools & IDEs** (VSCode, Eclipse, Microchip Studio, MPLAB, STM32CubeIDE, Keil uVision, Arduino IDE) | **Simulation & Platforms** (Proteus, LabVIEW) | **Version Control** (Git, GitHub) | **Operating Systems** (FreeRTOS , Embedded Linux) | **Others** (Data Structures, Algorithms, Networking Basics, Hardware Interfacing, IoT,Linux Architecture, Package Management (.deb/dpkg))

PROJECTS

IoT Health Monitoring System, *ATmega32, MQTT, Blynk, UART, ADC*

- Developed an end-to-end IoT health monitoring system using ATmega32 to track patient vitals in real-time.
- Interfaced temperature and heartbeat sensors via ADC, utilizing UART and VSPE for reliable serial communication.
- Integrated MQTT protocol to transmit sensor data to the Blynk platform for remote monitoring, triggering alarms upon critical thresholds. [GitHub Link 🔗]

Autonomous Fire-Fighting Robot, *Arduino, Sensors, Actuators*

- Designed and built an autonomous robot equipped with flame sensors for accurate fire detection and rapid response.
- Integrated an ultrasonic sensor and servo motor for dynamic obstacle avoidance and navigation.
- Utilized motor drivers to control DC motors for precise movement and water pump deployment.

AVR Embedded Systems Applications, *ATmega32, Embedded C*

- Developed modular Embedded C drivers for various AVR-based applications including a Real-Time Clock (RTC), Smart Safe, and Electronic Voting Machine (EVM).
- Implemented efficient hardware interfacing with keypads, LCDs, and EEPROM, utilizing timers and interrupts to optimize system performance. [GitHub Link 🔗]

C Console Applications (Management Systems), *C Programming*

- Programmed robust Bank and Employee Management systems in C, focusing on advanced data structures and dynamic memory allocation.
- Applied modular programming and efficient file handling techniques to ensure secure data storage and retrieval. [GitHub Link 🔗]

EXPERIENCE & VOLUNTEERING

- IEEE Al-Azhar Branch, Head of Embedded Systems** 2025 – Present | Cairo, Egypt
- Direct the embedded systems committee by managing the team, delegating technical tasks, and overseeing project execution .
- Develop and deliver instructional presentations and slides to effectively teach embedded systems concepts .
- Organize structured technical training tracks, including designing hands-on assignments and evaluating member progress .
- National Telecommunication Institute (NTI), IoT Trainee** 2026 | Online
Completed intensive training on IoT architectures, communication protocols, and system integration.
- Technological Development Programs with FA Model at FJ, IoT Instructor** 2025 | Online
Delivered technical sessions on IoT fundamentals and technological development to program participants.
- IEEE Al-Azhar Branch, Embedded Systems Instructor – Code Baker Camp** 2025 | Online
Instructed new members on the foundational concepts of embedded systems and PIC microcontrollers.
- IEEE Al-Azhar Branch, Arduino MCU Instructor – The Little Al-Khwarizmi Camp** 2025 | Online
Simplified complex technical concepts and taught Arduino programming to children and beginners.
- IEEE Al-Azhar Branch, Embedded Systems Member** 2025 | Cairo, Egypt
Participated in technical workshops and projects focusing on PIC microcontrollers and embedded C.
- IEEE Al-Azhar Branch, Python Member** 2024 | Cairo, Egypt
Developed foundational programming skills and problem-solving techniques using Python.

TRAININGS & ACHIEVEMENTS

- IoT Intensive Training Program, National Telecommunication Institute (NTI)** 2026 | Online
The training covered:
- Advanced IoT & Protocols: Deep dive into MQTT, system integration, and IoT architectures.
- Review & Implementation: Comprehensive review of Embedded C, ATmega32 drivers, and hardware interfacing.
- Specialized Technical Courses, Cisco, Harvard, and Mahara-Tech** 2025 | Online
The training covered:
- Networking Basics (Cisco): Completed a 40-hour track on networking fundamentals.
- Introduction to IoT (Harvard University): Explored embedded systems, sensors, and network connectivity.
- IoT Fundamentals (Mahara-Tech): Gained insights into modern IoT system designs and architectures.
- ARM Architecture: Mastered ARM Cortex-M basics [GitHub Link [↗](#)].
- Embedded Systems Diploma (230 Hours), With Eng. Mohamed Zaghloul** 2024 | Online
The training covered :
- Microcontroller Interfacing : Developed modular drivers for AVR peripherals [GitHub Link [↗](#)].
- Core Concepts : C/Embedded C, Data Structures, Algorithms, Testing, and Computer Architecture.
- System Design : RTOS, Bootloader basics, and Communication Protocols (USART, SPI, I2C).

HONORS & ACHIEVEMENTS

HackerRank
Achieved the 5-Star Gold Badge in C Programming.

Self-Learning
Built a solid foundation in Python (2022) and C++ (2023) through intensive independent study.

LANGUAGES

Arabic (Native) | **English** (Professional Working Proficiency) | **German** (Basic Knowledge)